

Haptics in Robotics: A Systematic Literature Review

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Abstract. Tactile sensing has emerged as a critical capability for robotic systems in contact-rich environments where vision and position control alone are insufficient. Following PRISMA 2020 guidelines, this review synthesises 19 studies on haptic and tactile sensing for robotic manipulation, covering sensor design, multimodal perception, teleoperation feedback, and learning-based control. High-resolution tactile sensors substantially improve grasp stability, slip detection, and in-hand manipulation. Embedding tactile signals into reinforcement learning and visual-tactile pre-training frameworks improves success rates and generalisation. Neuroscience findings on sensorimotor integration provide additional design principles for robotic haptic interfaces. Key challenges remain in sensor durability, computational cost, and sim-to-real transfer.

Keywords. haptics, tactile sensing, robotic manipulation, multimodal perception, contact-aware control, visuo-tactile learning

1. Introduction

The sense of touch plays a fundamental role in how humans perceive, explore, and interact with the physical world. Unlike vision or audition, tactile perception arises through direct physical contact and is inherently coupled with action and motor control, enabling object manipulation, tool use, and fine motor coordination [1–3]. Haptic technologies aim to recreate or substitute touch sensations, with applications in surgical robotics, teleoperation, rehabilitation, assistive devices, and social communication [4, 5]. Producing artificial touch that is natural, intuitive, and functionally effective remains a significant challenge.

Traditional robotic manipulation relied on visual sensing and position control, leading to failures in contact-rich environments. Modern systems combine vision, proprioception, and tactile sensing [6, 7]. Advances in electronic skin, optical tactile sensors, and flexible pressure arrays have enabled dexterous manipulation and teleoperation in unstructured environments [8, 9]. Integrating tactile with visual feedback improves robustness and generalisation [10, 11]. Given the growing role of robotics in healthcare from minimally invasive surgery to rehabilitation—this review synthesises neuroscience-informed and learning-based approaches to robotic haptics, identifying design principles and open challenges for researchers in this domain.

2. Methods

This review followed the PRISMA 2020 guidelines [12]. Relevant studies on haptics, tactile sensing, and robotic manipulation were identified through a structured search of scientific databases.

2.1. Search Strategy

Literature searches were conducted using PubMed, Web of Science, IEEE Xplore, ACM Digital Library, and Google Scholar, covering January 2000 to December 2025. Search terms included “haptics”, “tactile sensing”, “robotic manipulation”, “electronic skin”, “visuo-tactile”, “somatosensory”, and “sensorimotor integration”, adapted for each database. Only English-language full-text articles were considered.

2.2. Eligibility Criteria

Studies were included if they investigated tactile or haptic sensing in robotics, perception, or sensorimotor control. Both experimental studies and review papers were considered when they provided empirical results or clear analysis related to touch perception, tactile sensors, or haptic interaction.

2.3. Exclusion Criteria

Papers focusing only on vision, control algorithms without tactile feedback, or non-peer-reviewed material were excluded. Studies unrelated to touch perception, haptic systems, or robotic manipulation were not considered.

2.4. Study Selection and Synthesis

After removing duplicates, titles and abstracts were screened, followed by full-text review of relevant papers. Selected studies were grouped into thematic categories including tactile sensing technologies, multimodal perception, and contact-aware robotic manipulation. Due to methodological differences among studies, a qualitative synthesis was performed instead of a quantitative meta-analysis.

3. Results

3.1. Biological Principles Informing Robotic Haptic Design

A neurocognitive framework argued that haptic interfaces must align with human sensory, motor, and cognitive processes rather than optimise for physical fidelity alone [13].

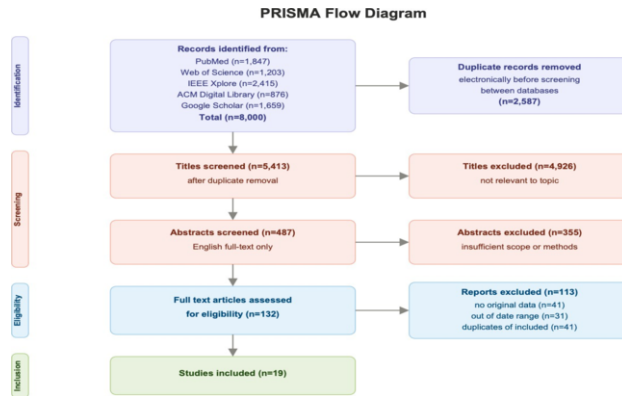


Figure 1: PRISMA flow diagram of study selection.

A user study identified 14 stable tactile experience categories linked to mechanoreceptor activation, providing a perception-grounded vocabulary for haptic device evaluation [14]. Cross-modal inhibition between tactile submodalities showed that co-activating different receptor channels can suppress intended percepts, constraining multimodal haptic design [15]. Motor commands were found to strongly modulate tactile spatial perception, with greater interference during active than passive movement [16]. Action observation recruits the same predictive sensorimotor mechanisms as execution, supporting temporally coherent haptic feedback in robotics [2]. A VR experiment confirmed that removing haptic feedback degrades eye–hand coordination and manipulation performance, establishing touch as constitutive of sensorimotor control [17].

3.2. Tactile Sensor Development for Robotic Hands

The GelSight sensor demonstrated significant improvements in grasp stability, slip detection, and object recognition via a vision-based deformable elastomer surface [18]. The F-TAC Hand achieved tactile coverage across approximately 70% of the hand surface at 0.1 mm resolution, substantially outperforming non-tactile systems in grasping, dynamic manipulation, and in-hand pose estimation across approximately 600 trials [19]. PolyTouch, a compact multi-modal finger combining camera-based tactile sensing, acoustic imaging, and peripheral vision, achieved a 20-fold longer lifetime than commercial sensors and outperformed haptic-oblivious control using tactile-diffusion policies [20].

3.3. Haptic Feedback in Teleoperation Systems

Peripheral nerve stimulation preserving somatotopic organisation and natural temporal encoding was shown to restore meaningful touch in prosthetic users, with adaptation through neural plasticity [21]. In robotic teleoperation, Tactile data alone was sufficient to reconstruct 3D object geometry and provide useful operator feedback without visual input, demonstrating tactile sensing as a primary channel in remote surgical [22].

3.4. Learning-Based Tactile Control for Robotic Manipulation

Tac2Motion, a reinforcement learning framework embedding tactile contact in the observation space with reward shaping, successfully transferred policies from simula-

tion to a real multi-fingered hand on contact-rich tasks [23]. Visual-tactile pre-training improved manipulation success by over 60% versus no pre-training and outperformed vision-only methods, establishing tactile representations as a first-class signal for data-driven control [24].

4. Discussion

Three themes characterise the literature. First, neural plausibility guides effective design: systems aligned with biological tactile encoding—somatotopic organisation and sensorimotor coherence—outperform those optimised for physical fidelity alone [13, 21]. Second, touch is indispensable; removing tactile feedback degrades control quality in ways vision cannot compensate [17, 19, 23]. Third, multi-modal learned representations are increasingly central, with tactile, visual, and acoustic fusion yielding more robust policies [20, 22, 24].

Limitations include sensor fragility and calibration demands [18, 19], sim-to-real gaps [23, 24], and insufficient translation of neuroscience findings into device specifications [14, 15]. These are especially critical in healthcare robotics where reliability and real-time performance are essential.

5. Conclusion

This review examined haptics in robotics across the neuroscience of touch and the engineering of tactile systems. Effective robotic haptics requires interfaces informed by biological encoding principles [13, 21], sensorimotor architectures respecting action–perception coupling [16, 17], and learning pipelines leveraging multi-modal tactile representations [20, 24]. The field is converging toward contact-aware systems for unstructured environments including clinical and rehabilitation settings. Future work should prioritise sensor durability, diverse tactile datasets, and closer alignment between neuroscience and haptic interface design.

References

- [1] Stoffregen T, Mantel B, Bardy B. The senses considered as one perceptual system. 2024.
- [2] Flanagan JR, Johansson RS. Action plans used in action observation. *Nature*. 2003;424(6950): 769-771.
- [3] Srinivasan MA, Basdogan C. Haptics in virtual environments: Taxonomy, research status, and challenges. *Computers and Graphics (Pergamon)*. 1997;21: 393-404.
- [4] Burdea G. Force and touch feedback for virtual reality. New York: John Wiley & Sons; 1996.
- [5] Cruz TPC. Haptic feedback in robot-assisted minimally invasive surgery. 2025.
- [6] Bicchi A, Kumar V. Robotic grasping and contact: A review. *Proceedings of the IEEE*. 2000;88(6): 348-358.
- [7] Dahiya RS, Metta G, Valle M, Sandini G. Tactile sensing—from humans to humanoids. *IEEE Transactions on Robotics*. 2010;26(1): 1-20.
- [8] Yuan W, Dong S, Adelson E. GelSight: High-resolution robot tactile sensors for estimating geometry and force. *Sensors*. 2017;17(12): 2762.
- [9] Rus D, Tolley MT. Design, fabrication and control of soft robots. *Nature*. 2015;521: 467-475.
- [10] Lee MA, et al. Making sense of vision and touch: Self-supervised learning of multimodal representations. *IEEE Robotics and Automation Letters*. 2019.
- [11] Calandra R, et al. More than a feeling: Learning to grasp and regrasp using vision and touch. *IEEE Robotics and Automation Letters*. 2018.

- [12] Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: An updated guideline for reporting systematic reviews. *BMJ*. 2021;372: n71.
- [13] Nisky I, Makin TR. A neurocognitive pathway for engineering artificial touch. *Science Advances*. 2024;10(51): eadq6290.
- [14] Obrist M, Seah S, Subramanian S. Talking about tactile experiences. In: *Conference on Human Factors in Computing Systems (CHI)*. 2013.
- [15] Cataldo A, Hagura N, Hyder YF, Haggard P. Touch inhibits touch: sanshool-induced paradoxical tingling reveals perceptual interaction between somatosensory submodalities. *Proceedings of the Royal Society B: Biological Sciences*. 2021;288(1943): 20202914.
- [16] Cataldo A, Dupin L, Dempsey-Jones H, Gomi H, Haggard P. Interplay of tactile and motor information in constructing spatial self-perception. *Current Biology*. 2022;32(6): 1301-1309.e3.
- [17] Lavoie E, Hebert JS, Chapman CS. How a lack of haptic feedback affects eye-hand coordination and embodiment in virtual reality. *Scientific Reports*. 2025;15(1): 25219.
- [18] Dong S, et al. High-resolution tactile sensing for reactive robotic manipulation [PhD thesis]. Massachusetts Institute of Technology; 2021.
- [19] Zhao Z, Li W, Li Y, Liu T, Li B, Wang M, et al. Embedding high-resolution touch across robotic hands enables adaptive human-like grasping. *Nature Machine Intelligence*. 2025;7(6): 889-900.
- [20] Zhao J, Kuppuswamy N, Feng S, Burchfiel B, Adelson E. Polytouch: A robust multi-modal tactile sensor for contact-rich manipulation using tactile-diffusion policies. In: *2025 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE; 2025. p. 104-110.
- [21] Bensmaia SJ, Tyler DJ, Micera S. Restoration of sensory information via bionic hands. *Nature Biomedical Engineering*. 2023;7(4): 443-455.
- [22] Giudici G, Bonzini AA, Coppola C, Althoefer K, Farkhatdinov I, Jamone L. Leveraging tactile sensing to render both haptic feedback and virtual reality 3d object reconstruction in robotic telemanipulation. *arXiv preprint arXiv:2412.02644*. 2024.
- [23] Kim Y, Rask CH, Sloth C. Tac2motion: Contact-aware reinforcement learning with tactile feedback for robotic hand manipulation. *arXiv preprint arXiv:2509.17812*. 2025.
- [24] Liu Q, Ye Q, Sun Z, Cui Y, Li G, Chen J. Masked visual-tactile pre-training for robot manipulation. In: *2024 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE; 2024. p. 13859-13875.